

Introduction to Artificial Intelligence (AI): definition, history, applications. AI agent overview: definition of rational agent, agent environment, agent structure.

Introduction

- ✓ Artificial Intelligence is concerned with the design of intelligence in an artificial device. The term was coined by John McCarthy in 1956.
- ✓ Intelligence is the ability to acquire, understand and apply the knowledge to achieve goals in the world.
- ✓ AI is the study of the mental faculties through the use of computational models
- ✓ AI is the study of intellectual/mental processes as computational processes.

Introduction contd.

- ✓ AI program will demonstrate a high level of intelligence to a degree that equals or exceeds the intelligence required of a human in performing some task.
- ✓ AI is unique, sharing borders with Mathematics, Computer Science, Philosophy, Psychology, Biology, Cognitive Science and many others.
- ✓ Although there is no clear definition of AI or even Intelligence, it can be described as an attempt to build machines that like humans can think and act, able to learn and use knowledge to solve problems on their own.

History of AI

Important research that laid the groundwork for AI:

- In 1931, Goedel laid the foundation of Theoretical Computer Science 1920-30s: He published the first universal formal language and showed that math itself is either flawed or allows for unprovable but true statements.
- In 1936, Turing reformulated Goedel's result and Church's extension thereof.
- In 1956, John McCarthy coined the term "Artificial Intelligence" as the topic of the Dartmouth Conference, the first conference devoted to the subject.
- In 1957, The General Problem Solver (GPS) demonstrated by Newell, Shaw & Simon
- In 1958, John McCarthy (MIT) invented the Lisp language.

History of AI contd.

- In 1959, Arthur Samuel (IBM) wrote the first game-playing program, for checkers, to achieve sufficient skill to challenge a world champion.
- In 1963, Ivan Sutherland's MIT dissertation on Sketchpad introduced the idea of interactive graphics into computing.
- In 1966, Ross Quillian (PhD dissertation, Carnegie Inst. of Technology; now CMU) demonstrated semantic nets
- In 1967, Dendral program (Edward Feigenbaum, Joshua Lederberg, Bruce Buchanan, Georgia Sutherland at Stanford) demonstrated to interpret mass spectra on organic chemical compounds. First successful knowledge-based program for scientific reasoning.
- In 1967, Doug Engelbart invented the mouse at SRI

History of AI contd.

- In 1968, Marvin Minsky & Seymour Papert publish Perceptrons, demonstrating limits of simple neural nets.
- In 1972, Prolog developed by Alain Colmerauer.
- In Mid 80's, Neural Networks become widely used with the Backpropagation algorithm (first described by Werbos in 1974).
- 1990, Major advances in all areas of AI, with significant demonstrations in machine learning, intelligent tutoring, case-based reasoning, multi-agent planning, scheduling, uncertain reasoning, data mining, natural language understanding and translation, vision, virtual reality, games, and other topics.
- In 1997, Deep Blue beats the World Chess Champion Kasparov
- In 2002, iRobot, founded by researchers at the MIT Artificial Intelligence Lab, introduced Roomba, a vacuum cleaning robot. By 2006, two million had been sold.

Foundations of AI

- ✓ Philosophy e.g., foundational issues (**can a machine think?**), issues of knowledge and believe, mutual knowledge.
- ✓ Psychology and Cognitive Science e.g., problem solving skills
- ✓ Neuro-Science e.g., brain architecture
- ✓ Computer Science and Engineering e.g., complexity theory, algorithms, logic and inference, programming languages, and system building.
- ✓ Mathematics and Physics e.g., statistical modeling, continuous mathematics,
- ✓ Statistical Physics, and Complex Systems.

Sub Areas of AI

1. **Game Playing** Deep Blue Chess program beat world champion Gary Kasparov
2. **Speech Recognition** PEGASUS spoken language interface to American Airlines' EAASY SABRE reservation system, which allows users to obtain flight information and make reservations over the telephone. The 1990s has seen significant advances in speech recognition so that limited systems are now successful.
3. **Computer Vision** Face recognition programs in use by banks, government, etc. The ALVINN system from CMU autonomously drove a van from Washington, D.C. to San Diego (all but 52 of 2,849 miles), averaging 63 mph day and night, and in all weather conditions. Handwriting recognition, electronics and manufacturing inspection, photo interpretation, baggage inspection, reverse engineering to automatically construct a 3D geometric model.

Sub Areas of AI contd.

4. Expert Systems Application-specific systems that rely on obtaining the knowledge of human experts in an area and programming that knowledge into a system.

- a) **Diagnostic Systems:** MYCIN system for diagnosing bacterial infections of the blood and suggesting treatments. Intellipath pathology diagnosis system (AMA approved). Pathfinder medical diagnosis system, which suggests tests and makes diagnoses. Whirlpool customer assistance center.
- b) **System Configuration:** DEC's XCON system for custom hardware configuration. Radiotherapy treatment planning.
- c) **Financial Decision Making:** Credit card companies, mortgage companies, banks, and the U.S. government employ AI systems to detect fraud and expedite financial transactions. For example, AMEX credit check
- d) **Classification Systems:** Put information into one of a fixed set of categories using several sources of information. E.g., financial decision making systems. NASA developed a system for classifying very faint areas in astronomical images into either stars or galaxies with very high accuracy by learning from human experts' classifications.

Sub Areas of AI contd.

- 5. **Mathematical Theorem Proving** Use inference methods to prove new theorems.
- 6. **Natural Language Understanding** AltaVista's translation of web pages. Translation of Caterpillar Truck manuals into 20 languages.
- 7. **Scheduling and Planning** Automatic scheduling for manufacturing. DARPA's DART system used in Desert Storm and Desert Shield operations to plan logistics of people and supplies. American Airlines rerouting contingency planner. European space agency planning and scheduling of spacecraft assembly, integration and verification.
- 8. **Artificial Neural Networks**
- 9. **Machine Learning**

Application of AI

AI algorithms have attracted close attention of researchers and have also been applied successfully to solve problems in engineering. Nevertheless, for large and complex problems, AI algorithms consume considerable computation time due to stochastic feature of the search approaches

- 1) *Business; financial strategies*
- 2) *Engineering: check design, offer suggestions to create new product, expert systems for all engineering problems*
- 3) *Manufacturing: assembly, inspection and maintenance*
- 4) *Medicine: monitoring, diagnosing*
- 5) *Education: in teaching*
- 6) *Fraud detection*
- 7) *Object identification*
- 8) *Information retrieval*
- 9) *Space shuttle scheduling*

Building AI Systems

- 1) Perception** Intelligent biological systems are physically embodied in the world and experience the world through their sensors (senses). For an autonomous vehicle, input might be images from a camera and range information from a rangefinder. For a medical diagnosis system, perception is the set of symptoms and test results that have been obtained and input to the system manually.
- 2) Reasoning** Inference, decision-making, classification from what is sensed and what the internal "model" is of the world. Might be a neural network, logical deduction system, Hidden Markov Model induction, heuristic searching a problem space, Bayes Network inference, genetic algorithms, etc. Includes areas of knowledge representation, problem solving, decision theory, planning, game theory, machine learning, uncertainty reasoning, etc.
- 3) Action** Biological systems interact within their environment by actuation, speech, etc. All behavior is centered around actions in the world. Examples include controlling the steering of a Mars rover or autonomous vehicle, or suggesting tests and making diagnoses for a medical diagnosis system. Includes areas of robot actuation, natural language generation, and speech synthesis.

Differences between strong AI and weak AI

Strong AI makes the bold claim that computers can be made to think on a level (at least) equal to humans.

Weak AI simply states that some "thinking-like" features can be added to computers to make them more useful tools... and this has already started to happen (witness expert systems, drive-by-wire cars and speech recognition software).

AI Problems: AI problems (speech recognition, NLP, vision, automatic programming, knowledge representation, etc.) can be paired with techniques (NN, search, Bayesian nets, production systems, etc.).

What's involved in Intelligence?

- Ability to interact with the world (speech, vision, motion, manipulation)
- Ability to model the world and to reason about it
- Ability to learn and to adapt
 - Perceive
 - Understand
 - Predict
 - Manipulate

Goals in AI

- To build systems that exhibit intelligent behavior
- To understand intelligence in order to model it

What is AI?

Hard Question?

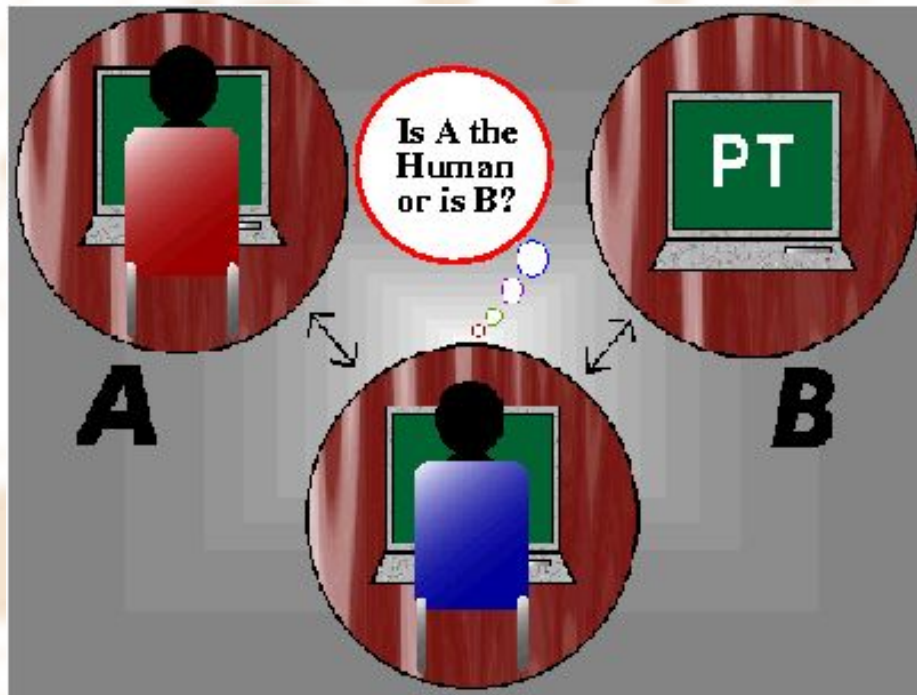
Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.	

What does modern AI do?

- ✓ Acting Humanly: Turing Test, not much pursued outside Loebner prize
 - Aeronautics: building pigeons that can fly so much like real pigeons that they can fool pigeons
 - Not reproducible, not amenable to mathematical analysis
- ✓ Thinking Humanly: Cognitive Science. How do humans think? How does the (human) brain work?
 - Neural networks are a (extremely simple so far) approximation
- ✓ Thinking Rationally: Logics, Formalization of knowledge and inference
 - You know the basics, we do some more, fairly widespread in modern AI
- ✓ Acting Rationally: How to make good action choices?
 - Contains Logics (one possible way to make intelligent decisions)
 - We are interested in making good choices in practice (e.g. in AlphaGo)

Acting Humanly

- Turing test: ultimate test for acting humanly
 - Computer and human both interrogated by judge
 - Computer passes test if judge can't tell the difference



Acting Humanly contd.

- Agent must:
 - Have command of language
 - Have wide range of knowledge
 - Demonstrate human traits (humor, emotion)
 - Be able to reason
 - Be able to learn
- [Loebner prize](#) competition is modern version of Turing Test
 - Example: [Alice](#), Loebner prize winner for 2000 and 2001

Wikipedia

Cognition (/kɒɡˈnɪ(ə)n/ (listen)) refers to "the mental action or process of acquiring knowledge and understanding through thought, experience, and the senses".^[2] It encompasses many aspects of intellectual functions and processes such as: perception, attention, the formation of knowledge, memory and working memory, judgment and evaluation, reasoning and "computation", problem solving and decision making, comprehension and production of language. Cognitive processes use existing knowledge and discover new knowledge.

Cognitive processes are analyzed from different perspectives within different contexts, notably in the fields of linguistics, musicology, anesthesia, neuroscience, psychiatry, psychology, education, philosophy, anthropology, biology, systemics, logic, and computer science.^[3]

These and other different approaches to the analysis of cognition are synthesised in the developing field of cognitive science, a progressively autonomous academic discipline.

Thinking Humanly: Cognitive Science

- 1960s "**cognitive revolution**": information-processing psychology replaced prevailing orthodoxy of behaviorism
- Requires scientific theories of internal activities of the brain
- What level of abstraction? "Knowledge" or "circuits"?
- How to validate? Requires
 - (1) Predicting and testing behaviour of human subjects (top-down)
 - (2) Direct identification from neurological data (bottom-up)
- Both approaches (roughly, Cognitive Science and Cognitive Neuroscience) are now distinct from AI

Thinking Rationally: laws of thought

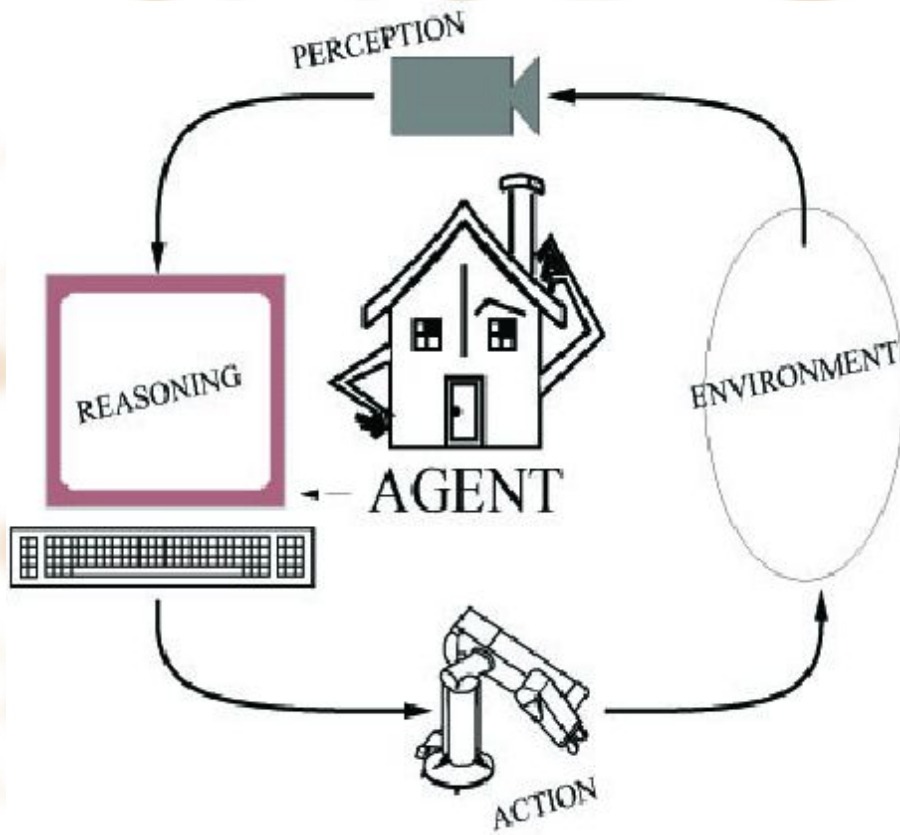
- What are correct arguments or thought processes?
- Provided foundation of much of AI
- Not all intelligent behavior controlled by logic
- What is our goal? What is the purpose of thinking?
- Normative (or prescriptive) rather than descriptive
- Aristotle: What are correct arguments/thought processes?
- Several Greek schools developed various forms of logic: notation and rules of derivation for thoughts; may not have proceeded to the idea of mechanization
- Direct line through mathematics and philosophy to modern AI
- Problems:
 - 1) Not all intelligent behavior is mediated by logical deliberation
 - 2) What is the purpose of thinking? What thoughts should I have out of all the thoughts (logical or otherwise) that I could have?

Acting Rationally

Rational behavior: doing the right thing

- The right thing: that which is expected to maximize goal achievement, given the available information
- Doesn't necessarily involve thinking—e.g., blinking reflex—but
- Thinking should be in the service of rational action
- Aristotle (Nicomachean Ethics):
 - Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good
 - Act to achieve goals, given set of beliefs
- Rational behavior is doing the “right thing”
 - Thing which expects to maximize goal achievement

Components of an AI System



An **agent** **perceives** its environment through **sensors** and **acts** on the environment through **actuators**.

Human: sensors are eyes, ears, actuators (effectors) are hands, legs, mouth.

Robot: sensors are cameras, sonar, lasers, ladar, bump, effectors are grippers, manipulators, motors

The agent's behavior is described by its function that maps percept to action.

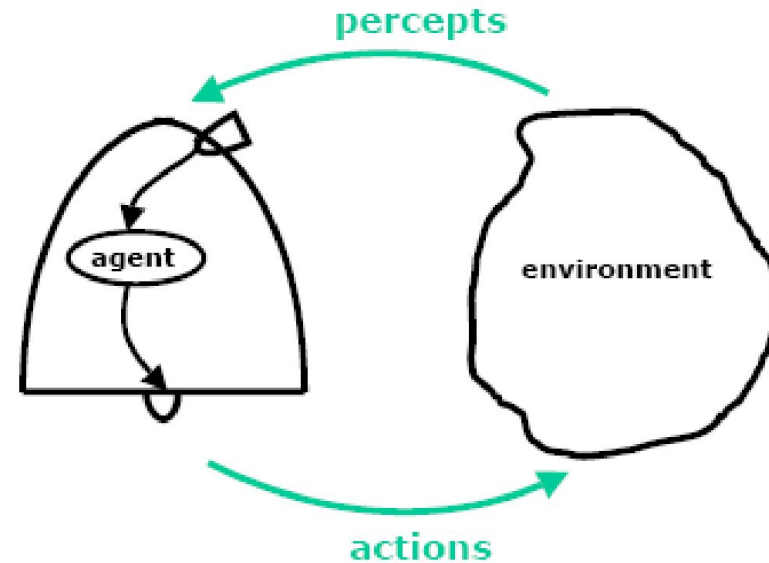
An Agent

‘Anything’ that can gather information about its environment and take action based on that information.



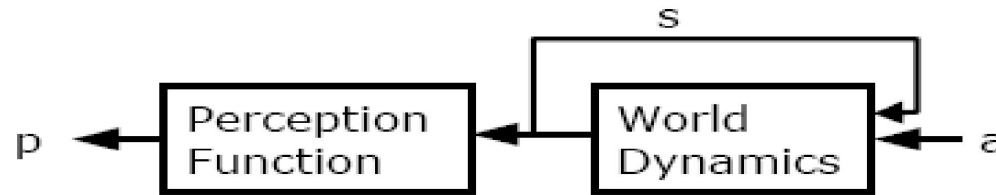
The Agent Environment

- What all do we need to specify?
 - The action space
 - The percept space
 - The environment as a string of mappings from the action space to the percept space.
- A rational agent acts in its environment, according to what it has perceived, in order to maximize its expected performance measure.



The World Model

- Perception function
- World dynamics / State transition function



- Utility function- how does the agent know what constitutes “good” or “bad” behavior

Agent and Environment

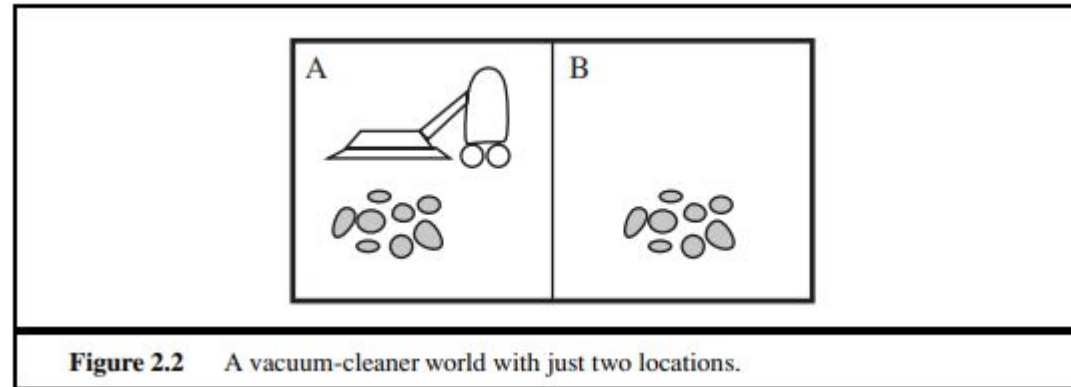
Percept: We use the term percept to refer to the agent's perceptual inputs at any given instant.

Percept Sequence: An agent's percept sequence is the complete history of everything the agent has ever perceived.

Agent function: Mathematically speaking, we say that an agent's behavior is described by the agent function that maps any given percept sequence to an action.

Agent program: Internally, the agent function for an artificial agent will be implemented by an agent program. It is important to keep these two ideas distinct. The agent function is an abstract mathematical description; the agent program is a concrete implementation, running on the agent architecture.

Example



Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

(must go through books)

Task Environment

- Use PEAS to describe task environment
 - Performance measure
 - Environment
 - Actuators
 - Sensors

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

Properties of Task Environments

- Accessible/Inaccessible
- Deterministic/Non-deterministic
- Static/Dynamic
- Discrete/Continuous
- E.g. Driving a car, a game of Chinese-checkers

(must do homework on this section)

Environment Examples



Fully observable vs. partially observable

Deterministic vs. stochastic / strategic

Episodic vs. sequential

Static vs. dynamic

Discrete vs. continuous

Single agent vs. multi agent

Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Chess with a clock	Fully	Strategic	Sequential	Semi	Discrete	Multi
Chess without a clock	Fully	Strategic	Sequential	Static	Discrete	Multi
Poker	Partial	Strategic	Sequential	Static	Discrete	Multi
Backgammon	Fully	Stochastic	Sequential	Static	Discrete	Multi
Taxi driving	Partial	Stochastic	Sequential	Dynamic	Continuous	Multi
Medical diagnosis	Partial	Stochastic	Episodic	Static	Continuous	Single
Image analysis	Fully	Deterministic	Episodic	Semi	Discrete	Single
Robot part picking	Fully	Deterministic	Episodic	Semi	Discrete	Single
Interactive English tutor	Partial	Stochastic	Sequential	Dynamic	Discrete	Multi

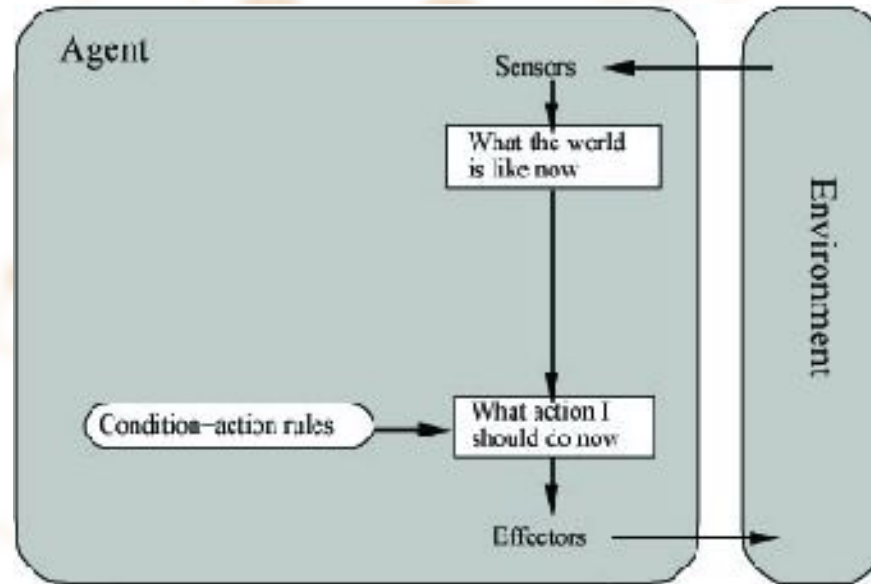
Agent Types

- Types of agents (increasing in generality and ability to handle complex environments)
 - Simple reflex agents
 - Reflex agents with state
 - Goal-based agents
 - Utility-based agents
 - Learning agent

Simple Reflex Agent

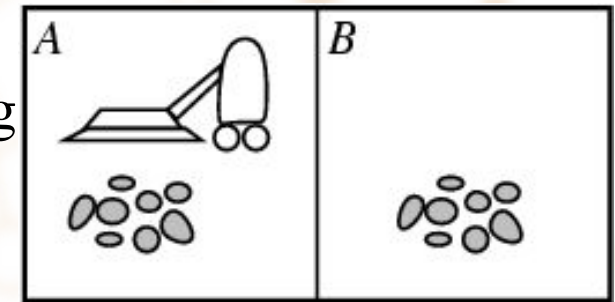
- Use simple “if then” rules
- Can be short sighted

```
SimpleReflexAgent(percept)
  state = InterpretInput(percept)
  rule = RuleMatch(state, rules)
  action = RuleAction(rule)
  Return action
```



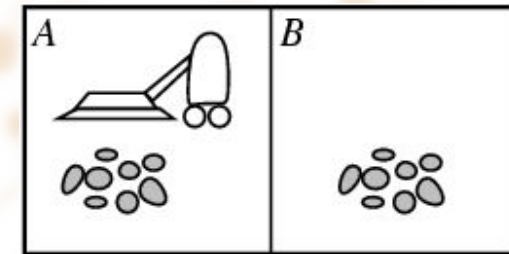
Example: Vacuum Agent

- Performance?
 - 1 point for each square cleaned in time T ?
 - #clean squares per time step - #moves per time step?
- Environment: vacuum, dirt, multiple areas defined by square regions
- Actions: left, right, suck, idle
- Sensors: location and contents
 - [A, dirty]
- Rational is not omniscient
 - Environment may be partially observable
- Rational is not clairvoyant
 - Environment may be stochastic
- Thus Rational is not always successful



Reflex Vacuum Agent

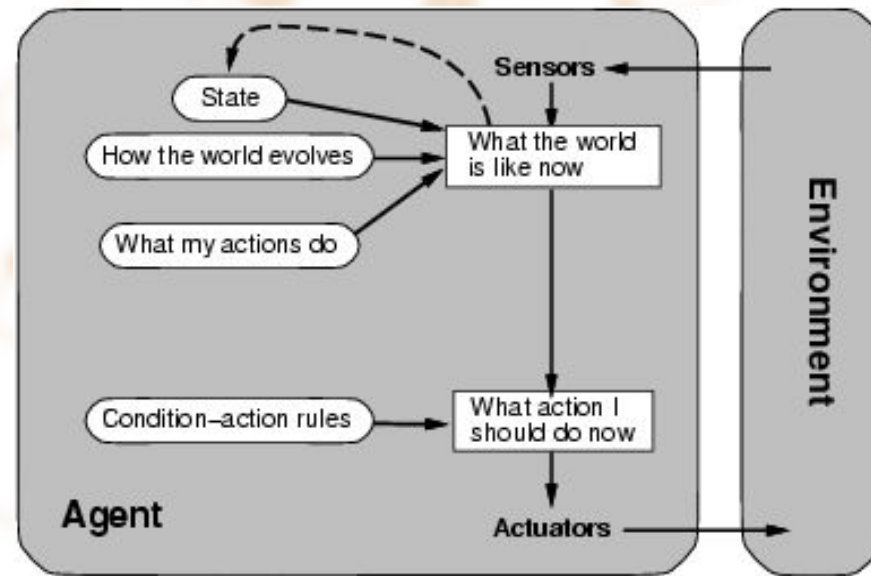
- If status=Dirty then return Suck
else if location=A then return Right
else if location=B then return Left



Reflex Agent With State

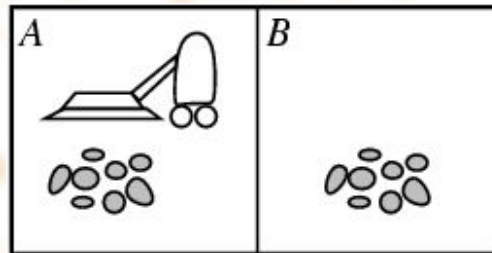
- Store previously-observed information
- Can reason about unobserved aspects of current state

```
ReflexAgentWithState(percept)
  state = UpdateDate(state,action,percept)
  rule  = RuleMatch(state, rules)
  action = RuleAction(rule)
  Return action
```



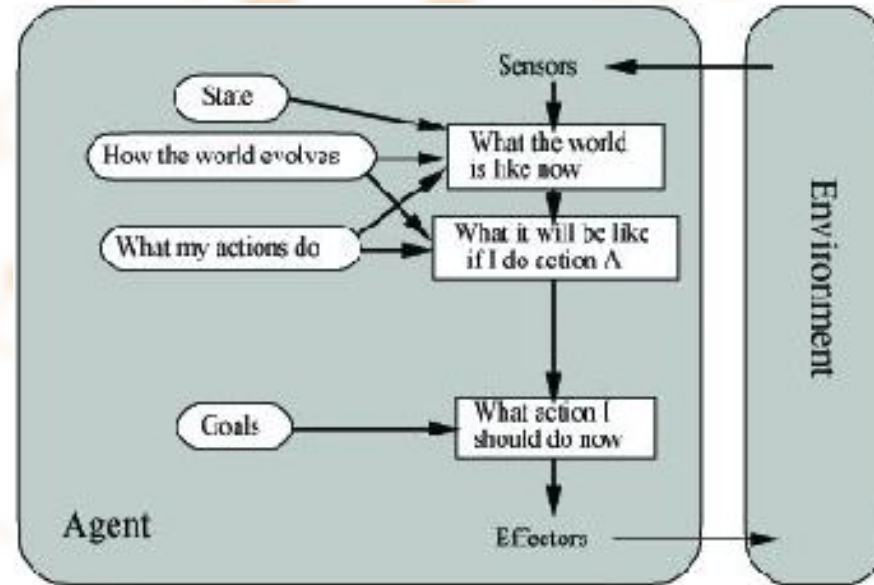
Reflex Vacuum Agent

- If status=Dirty then Suck
else if have not visited other square in >3 time units, go there



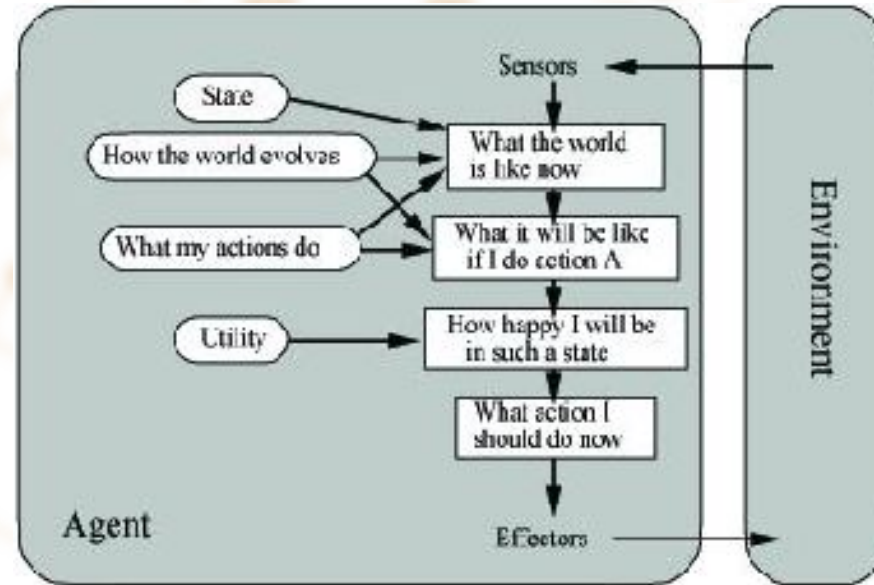
Goal-Based Agents

- Goal reflects desires of agents
- May project actions to see if consistent with goals
- Takes time, world may change during reasoning

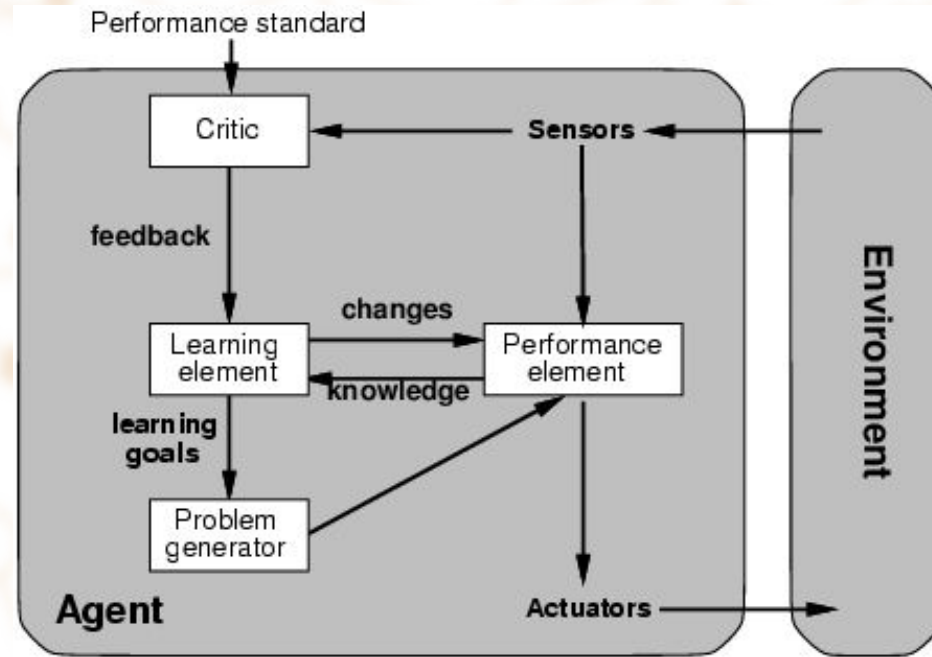


Utility-Based Agents

- Evaluation function to measure utility $f(\text{state}) \rightarrow \text{value}$
- Useful for evaluating competing goals



Learning Agents





thank
you